

Critical First Sign Failure of Even Cumulants for the Blume–Capel Single-Site Law

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Abstract

For the symmetric three-point law with mass $q/2$ at each of -1 and 1 and mass $1 - q$ at 0 , let κ_j be its cumulants. In the Blume–Capel parametrization

$$q = \frac{2e^{-\Delta}}{1 + 2e^{-\Delta}}, \quad \Delta > \log 2,$$

we determine the critical scale of the first even cumulant that violates the alternating sign pattern. If

$$m_*(\Delta) = \min\{m \geq 2 : (-1)^{m+1} \kappa_{2m} < 0\},$$

then the minimum exists for every $\Delta > \log 2$ and

$$\lim_{\varepsilon \downarrow 0} \sqrt{\varepsilon} m_*(\log 2 + \varepsilon) = \frac{\pi^2}{4\sqrt{2}}.$$

The proof uses an exact, absolutely convergent sum over all zeros of the moment-generating function. It also localizes m_* to fewer than two lattice steps above $\pi/[4 \arctan(a/\pi)]$, where $a = \operatorname{arcosh}(e^\Delta/2)$. This uniform control of every preceding order is the feature not supplied by fixed-parameter, high-order asymptotics.

1 Statement and context

The Blume–Capel model is the spin-one system originating in the works of Blume and Capel [1, 2]. A recent analysis of its Ursell functions identifies $\Delta = \log 2$ as the threshold for an alternating-sign property and proves that for each fixed $\Delta > \log 2$ the property fails at some even order [3, Proposition 4]. The present question is finer: as Δ approaches the threshold, at which *first* order does the strict failure occur? The fixed-parameter nearest-zero estimate in [3, equations (8)–(9)] does not by itself control all lower orders in this coupled limit.

Let X have distribution

$$\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = \frac{q}{2}, \quad \mathbb{P}(X = 0) = 1 - q,$$

where $0 < q < 1/2$. Its moment-generating function and cumulants are

$$M_q(z) = \mathbb{E}e^{zX} = 1 - q + q \cosh z, \quad \kappa_j(q) = \left. \frac{d^j}{dz^j} \log M_q(z) \right|_{z=0}.$$

Here and below, the logarithm means the analytic germ at the point where it is differentiated. For $\Delta > \log 2$, set

$$q = q(\Delta) = \frac{2}{e^\Delta + 2}, \quad a = a(\Delta) = \operatorname{arcosh}(e^\Delta/2) > 0.$$

Then $q = 1/(1 + \cosh a)$. Define

$$S_m(a) = (-1)^{m+1} \kappa_{2m}(q), \quad m_*(a) = \min\{m \in \mathbb{Z} : m \geq 2, S_m(a) < 0\}.$$

A zero value is therefore not counted as a failure.

Our quantitative form of the result is as follows.

Theorem 1. *For every $a > 0$, the defining set for $m_*(a)$ is nonempty. Put*

$$\beta = \arctan(a/\pi), \quad t(a) = \frac{\pi}{4\beta}.$$

For all sufficiently small $a > 0$,

$$t(a) < m_*(a) \leq \lceil t(a) + 1 \rceil < t(a) + 2. \quad (1)$$

Consequently,

$$\lim_{a \downarrow 0} a m_*(a) = \frac{\pi^2}{4}.$$

2 An exact zero sum

The uniformity needed for Theorem 1 comes from retaining all zeros rather than truncating to the nearest pair.

Proposition 2 (Exact sign formula). *For $a > 0$ and every integer $m \geq 1$, define*

$$\beta_k = \arctan\left(\frac{a}{\pi(2k+1)}\right), \quad k \geq 0.$$

Then

$$S_m(a) = 4(2m-1)! \sum_{k=0}^{\infty} \frac{\cos(2m\beta_k)}{(a^2 + \pi^2(2k+1)^2)^m}. \quad (2)$$

The series is absolutely convergent.

Proof. The elementary factorization

$$M_q(z) = \frac{\cosh z + \cosh a}{1 + \cosh a} = \frac{\cosh((z+a)/2) \cosh((z-a)/2)}{\cosh^2(a/2)}$$

shows, on writing $f(w) = \log \cosh w$, that

$$\kappa_{2m}(q) = 2^{1-2m} f^{(2m)}(a/2). \quad (3)$$

Euler's product

$$\cosh w = \prod_{k=0}^{\infty} \left(1 + \frac{4w^2}{\pi^2(2k+1)^2}\right)$$

may be differentiated termwise after the zeros have been paired. For derivatives of order at least two, the resulting series is locally normally convergent away from the zeros. With $y_k = \pi(2k+1)/2$, it gives

$$f^{(2m)}(x) = -(2m-1)! \sum_{k=0}^{\infty} ((x - iy_k)^{-2m} + (x + iy_k)^{-2m}). \quad (4)$$

At $x = a/2$, the argument of $a + i\pi(2k + 1)$ is $\pi/2 - \beta_k$. Substitution of (4) into (3), including the powers of two and the two conjugate terms, yields

$$\kappa_{2m}(q) = 4(-1)^{m+1}(2m-1)! \sum_{k=0}^{\infty} \frac{\cos(2m\beta_k)}{(a^2 + \pi^2(2k+1)^2)^m}.$$

This is (2). Absolute convergence follows by comparison with $\sum_{k \geq 0} (2k+1)^{-2m}$. $\square$

We first use the same formula to justify that m_* is defined throughout the stated parameter domain.

Lemma 3. *For every fixed $a > 0$, there are infinitely many integers m such that $S_m(a) < 0$.*

Proof. Normalize the summands in (2) by the positive weight of the $k = 0$ term and set

$$\rho_k = \frac{a^2 + \pi^2}{a^2 + \pi^2(2k+1)^2} \quad (k \geq 1).$$

Since $\rho_k \leq \rho_1 < 1$ and $\sum_{k \geq 1} \rho_k < \infty$,

$$\sum_{k \geq 1} \rho_k^m \leq \rho_1^{m-1} \sum_{k \geq 1} \rho_k \rightarrow 0. \quad (5)$$

There are infinitely many m for which $\cos(2m\beta_0) \leq -1/2$. Indeed, this follows from density of irrational rotations if β_0/π is irrational. If $\beta_0/\pi = p/r$ in lowest terms, then $0 < p < r/2$, so $r \geq 3$; the residues $mp \pmod{r}$ run through all residue classes, including one in the closed middle third, and this recurs periodically. Along either subsequence the negative $k = 0$ term has normalized magnitude at least $1/2$, while (5) tends to zero. Thus the leading term eventually dominates and $S_m(a) < 0$. $\square$

3 Localization of the first failure

Proof of Theorem 1. Existence follows from Lemma 3. Since $0 < \beta_k \leq \beta_0 = \beta$, with strict inequality for $k \geq 1$, an integer $m \leq t(a)$ satisfies

$$0 < 2m\beta_k \leq \frac{\pi}{2}.$$

Every summand in (2) is then nonnegative. The full sum is strictly positive: if equality occurs for the $k = 0$ angle, all $k \geq 1$ angles remain strictly below $\pi/2$. Hence

$$S_m(a) > 0 \quad \text{for every integer } m \leq t(a), \quad (6)$$

which proves $m_*(a) > t(a)$ and also handles the strict-zero convention.

For the upper bound let

$$n = n(a) = \lceil t(a) + 1 \rceil.$$

Then

$$\frac{\pi}{2} + 2\beta \leq 2n\beta < \frac{\pi}{2} + 4\beta. \quad (7)$$

For sufficiently small a , in particular when $\beta < \pi/8$, this interval lies in $(\pi/2, \pi)$, and therefore

$$\cos(2n\beta) \leq -\sin(2\beta). \quad (8)$$

It remains to control all the other zeros uniformly. For $a \leq \pi$ and $m \geq 1$, their total absolute weight relative to the $k = 0$ weight is

$$\begin{aligned} R_m(a) &:= \sum_{k=1}^{\infty} \left(\frac{a^2 + \pi^2}{a^2 + \pi^2(2k+1)^2} \right)^m \\ &\leq \left(\frac{2}{9} \right)^{m-1} \sum_{k=1}^{\infty} \frac{2}{(2k+1)^2} = \left(\frac{2}{9} \right)^{m-1} \left(\frac{\pi^2}{4} - 2 \right). \end{aligned} \quad (9)$$

Here one factor in each summand is bounded by $2/(2k+1)^2$, while every remaining factor is at most $2/9$. Since $n \geq \pi/(4\beta) + 1$, the last expression in (9) at $m = n$ is $o(\beta)$ as $\beta \downarrow 0$, whereas $\sin(2\beta) \sim 2\beta$. Thus, for all sufficiently small a ,

$$R_n(a) < \sin(2\beta).$$

Together with (2) and (8), this implies $S_n(a) < 0$. Combining this with (6) gives

$$t(a) < m_*(a) \leq n = \lceil t(a) + 1 \rceil < t(a) + 2,$$

which is (1).

Finally, $t(a) \rightarrow \infty$ as $a \downarrow 0$, so the localization yields $\beta m_*(a) \rightarrow \pi/4$. Since $\beta = \arctan(a/\pi) \sim a/\pi$, we obtain $a m_*(a) \rightarrow \pi^2/4$. $\square$

Corollary 4 (Critical Blume–Capel normalization). *With $m_*(\Delta)$ defined above,*

$$\lim_{\varepsilon \downarrow 0} \sqrt{\varepsilon} m_*(\log 2 + \varepsilon) = \frac{\pi^2}{4\sqrt{2}}.$$

Proof. For $\Delta = \log 2 + \varepsilon$, the definition of a gives $\cosh a = e^\varepsilon$. The expansions $e^\varepsilon = 1 + \varepsilon + O(\varepsilon^2)$ and $\cosh a = 1 + a^2/2 + O(a^4)$ imply $a/\sqrt{\varepsilon} \rightarrow \sqrt{2}$. The result follows from Theorem 1. $\square$

4 Exact computational check and scope

The proof above is analytic and does not depend on computation. As a check, consider the rational family $q_N = N/(2N+1)$ and write $D = 2N+1$. The moment–cumulant recurrence has the integer form

$$\begin{aligned} \kappa_{2m} &= \frac{A_m}{D^m}, \quad A_1 = N, \\ A_m &= N \left(D^{m-1} - \sum_{j=1}^{m-1} \binom{2m-1}{2j-1} A_j D^{m-1-j} \right). \end{aligned}$$

It determines every sign without floating-point arithmetic. Independent rational-arithmetic evaluation gives, for example,

N	20	100	1000	2000	10000	100000
m_*	8	18	56	79	175	552

and verifies all preceding signs in each case. These finite checks are corroboration only; they are not used to pass to the limit.

The result concerns the one-variable cumulants of the single-site law and does not assert a classification of general graph Ursell functions. Our literature comparison is deliberately bounded: Yu’s fixed-parameter result is the closest directly relevant statement located [3], but this comparison is not a claim of priority.

Computational and AI assistance. OpenAI Codex assisted with algebraic exploration, exact recurrence checks, proof auditing, and preparation of the manuscript. The recurrence was run with integer arithmetic and separately checked with exact rational arithmetic. A separate model context rederived the zero-sum identity and audited the proof; this is an internal verification step, not independent peer review or a formal proof. No numerical computation is used as evidence for the theorem.

References

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